

Active inference

A gentle introduction from neuroscience to embodied agents

Alexander Shaw

CPNS Lab · University of Exeter

a.d.shaw@exeter.ac.uk | cpnslab.com

definition, concept, equations, drones, robots and world models.

What this talk is trying to do

Make the active inference framework legible:

- Define the core objects: states, observations, beliefs, actions and policies.
- Build the equations slowly rather than treating them as slogans.
- Show how perception and action are coupled.

Active inference in brief

An active inference agent maintains a **generative model** of the causes of its sensory data, updates **beliefs** to explain what it observes and selects **actions** that make future observations both preferred and informative.

Core intuition

AI usually asks: *what action maximises reward?*

Active Inference asks: *what should I believe, what should I sample next and what kind of world am I trying to keep myself in?*

Where it came from

- The roots are in the **Bayesian brain**: perception as inference under uncertainty (Knill and Pouget 2004).
- Predictive coding made this computational: cortical hierarchies can be read as prediction and prediction-error systems (Rao and Ballard 1999).
- The Free Energy Principle proposed a broader account of perception, action and learning in self-organising systems (Friston 2010).
- Active inference turns this into a process theory for agents that infer hidden causes and act to resolve uncertainty (Friston et al. 2016; Da Costa et al. 2020).

Difference to the Bayesian brain:

BB: *the brain predicts the world*

ActInf: *the organism acts to sample and maintain the world it predicts.*

The problem it unifies

Perception

Infer hidden causes from sensory data.

Action

Change sensory data by moving through the world.

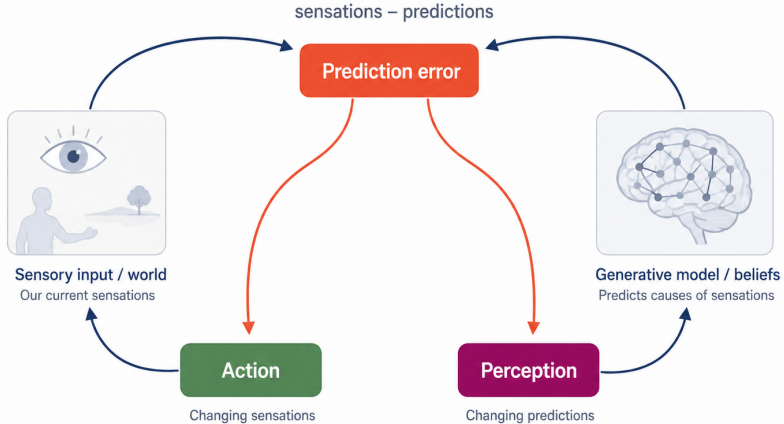
Learning

Change the model so future inference gets better.

beliefs $\leftrightarrow$ observations $\leftrightarrow$ actions

Active inference is useful because these are not really separate loops in embodied systems.

Making our own sensations



Notation: the objects we need

o	Observations or sensory data. These are what the agent receives.
s	Hidden states. These are the latent causes the agent must infer.
a	Actions. These change the world or the agent's sensory sampling.
π	A policy. Usually a sequence or rule for action over time.
$p(\cdot)$	The generative model: what the agent believes about how hidden states generate observations.
$q(\cdot)$	The approximate posterior: the agent's current tractable belief distribution.
$\mathcal{C}$	Preferences over outcomes: the observations the agent expects or seeks to occupy.

The generative model

$$p(o, s) = p(o | s)p(s)$$

$p(s)$

Prior belief about hidden states before the current observation.

$p(o | s)$

Likelihood: how likely each observation would be if a hidden state were true.

$p(o, s)$

Joint probability assigned by the model to observations and their causes.

In neuroscience this can mean a model of hidden causes of sensation.

In robotics it can mean a model of where the body, obstacles or target may be.

Bayesian perception

$$p(s \mid o) = \frac{p(o|s)p(s)}{p(o)}$$

- $p(s \mid o)$ is the posterior: belief about hidden states after observing o .
- $p(o \mid s)$ says what observations each hidden state predicts.
- $p(s)$ carries prior expectations.
- $p(o)$ is model evidence. It normalises the posterior but is often hard to compute.

The practical problem

Real agents rarely compute the exact posterior, so active inference uses a tractable approximation $q(s)$.

Variational inference: replace the impossible posterior

$$q(s) \approx p(s \mid o)$$

- $p(s \mid o)$ is the true posterior the agent would like to have.
- $q(s)$ is a simpler distribution the agent can update.
- The question becomes: how should $q(s)$ be changed so it becomes a good explanation of o ?

Answer

Minimise variational free energy, a tractable objective that scores how well beliefs explain observations under the model.

Variational free energy: the core equation

$$F[q] = \mathbb{E}_{q(s)} [\ln q(s) - \ln p(o, s)]$$

$F[q]$

A score for the agent's current beliefs $q(s)$. Lower is better.

$\mathbb{E}_{q(s)}$

Average over possible hidden states, weighted by current belief.
More likely states count more.

$\ln q(s)$

The log probability of each hidden-state hypothesis under the agent's current beliefs.

$\ln p(o, s)$

The log probability that the model assigns to the observation and its hidden cause together.

Minimising F updates beliefs so that the agent's hypotheses become better explanations of observed data.

Why free energy is useful

$$F[q] = D_{\text{KL}}(q(s) \parallel p(s \mid o)) - \ln p(o)$$

- $D_{\text{KL}}(q \parallel p)$ is a divergence between approximate and true posterior beliefs.
- $-\ln p(o)$ is surprise or negative model evidence.
- Because $D_{\text{KL}} \geq 0$, free energy is an upper bound on surprise*.

*Technically, free energy sits above surprise mathematically, but the important idea is simpler: it gives the agent something it can optimise.

Interpretation

The agent cannot usually minimise surprise directly because model evidence is hard to compute. It can minimise free energy instead.

Another useful decomposition

$$F[q] = \underbrace{D_{\text{KL}}(q(s) \| p(s))}_{\text{complexity}} - \underbrace{\mathbb{E}_{q(s)}[\ln p(o | s)]}_{\text{accuracy}}$$

Complexity

How much posterior beliefs depart from priors. This penalises over-complicated explanations.

Accuracy

How well hidden-state beliefs predict the actual observations.

Minimising F

Favour explanations that fit the data but do not move further from prior structure than needed.

Predictive coding as a special case

$$F \approx \frac{1}{2} \varepsilon^T \Pi \varepsilon + \text{complexity terms}$$

ε

Prediction error: the difference between what was observed and what was predicted.

Π

Precision: confidence or inverse variance assigned to prediction errors.

$\varepsilon^T \Pi \varepsilon$

Precision-weighted prediction error. Errors matter more when the system expects high reliability.

This is the bridge to cortical predictive coding and to much of CPNS computational psychiatry.

Action: the active part of active inference

Perception

Change beliefs so predictions match sensations.

$$q(s) \leftarrow \arg \min_q F[q]$$

Action

Change sensations so they become consistent with predictions and preferences.

$$a \leftarrow \arg \min_a F$$

Simple example

If I expect my hand to be on the cup, I can reduce prediction error by updating my belief, moving my hand, or both.

Preferences: what outcomes should the agent occupy?

$$p(o \mid C)$$

- C encodes prior preferences: which observations are desirable, viable or expected for the agent.
- $p(o \mid C)$ assigns probability to outcomes according to those preferences.
- Preferred observations have higher probability; avoided observations have lower probability.
- This is not a reward delivered after an action. It is a prior belief about the kinds of outcomes the agent should occupy.

In active inference, agents act to make preferred observations more probable.

Policies: imagined courses of action

$$\pi = (a_1, a_2, \dots, a_T)$$

- A policy π is a possible sequence of future actions.
- The agent uses its generative model to imagine what each policy would make it observe.
- Policies are scored by their expected free energy.
- Good policies are expected to produce outcomes that are both preferred and informative.

Action selection becomes a form of prospective inference: which policy will make the future most expected under the model?

Expected free energy: choosing what to do next

$$G(\pi) = \mathbb{E}_{q(o,s|\pi)} [\ln q(s | \pi) - \ln p(o, s | C)]$$

$G(\pi)$

A score for a possible policy π . Lower is better.

π

A possible sequence of future actions.

$q(o, s | \pi)$

What the agent predicts it would observe, and what hidden states it might enter, if it followed policy π .

$\mathbb{E}_{q(o,s|\pi)}$

Average over those imagined futures. More likely futures count more.

Expected free energy: what is being scored?

$$\ln q(s \mid \pi) - \ln p(o, s \mid C)$$

$$\ln q(s \mid \pi)$$

The agent's predicted belief about hidden states if it follows policy π .

$$p(o, s \mid C)$$

The model's probability of future observations and hidden states, including preferred outcomes C .

$$\ln p(o, s \mid C)$$

How well that imagined future fits the model and the agent's preferences.

Minimise G

Choose policies expected to produce futures that are preferred and informative.

In plain terms: the agent imagines possible futures, scores them, then acts to make the best expected future more likely.

Expected free energy: what it balances

$$G(\pi) \approx \underbrace{\mathbb{E}_{q(o|\pi)}[-\ln p(o | C)]}_{\text{risk}} + \underbrace{\mathbb{E}_{q(s|\pi)}[H[p(o | s)]]}_{\text{ambiguity}} - \underbrace{\mathbb{E}_{q(o|\pi)} D_{\text{KL}}(q(s | o, \pi) \| q(s | \pi))}_{\text{epistemic value}}$$

Risk

Expected mismatch between future observations and preferred observations.

Ambiguity

Expected uncertainty in the mapping from hidden states to observations.

Epistemic value

Expected information gain about hidden states. It is subtracted, so informative policies are favoured.

Policy selection

$$q(\pi) = \sigma(-\gamma G(\pi))$$

$q(\pi)$

Posterior belief that a policy should be pursued.

σ

Softmax function. Policies with lower G get higher probability.

γ

Policy precision or inverse temperature. Higher γ makes selection more decisive.

Important

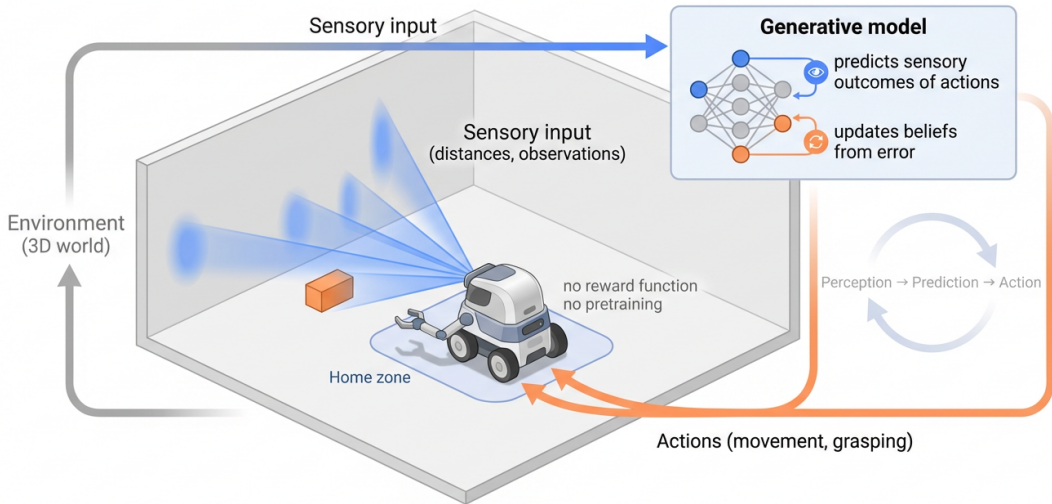
Action selection is also inference: the agent infers which policy it is likely to be pursuing.

The active inference loop

1. The world generates observations o_t .
2. The agent updates beliefs $q(s_t)$ by minimising variational free energy.
3. It predicts future observations under policies π .
4. It scores policies using expected free energy $G(\pi)$.
5. It acts, which changes the next observation.

observe $\rightarrow$ infer $\rightarrow$ imagine $\rightarrow$ select $\rightarrow$ act $\rightarrow$ observe

Behaviour emerges by reducing prediction error



Why this became interesting beyond neuroscience

- It gives a common language for **world modelling**, **uncertainty**, **information seeking** and **action**.
- It treats exploration as intrinsic to belief updating rather than as an extra bonus term.
- It naturally handles partial observability: agents act from beliefs, not privileged truth states.
- It scales conceptually from brains to embodied agents, though implementations still require careful engineering.

What active inference is not

- It is not a magic replacement for control theory, reinforcement learning or engineering.
- It is not one algorithm. It is a family of models and process theories.
- It does not remove implementation choices: state spaces, likelihoods, priors, precision and policy sets matter.
- It should be judged by whether the generative model, inference scheme and behaviour are useful for the problem.

A useful standard

Can the agent keep coherent beliefs, act adaptively under uncertainty and reveal why it acted as it did?

Why did you fiddle with it, Alex?

Standard active inference implementations often still compress behaviour into:

- one dominant posterior,
- one policy distribution,
- one objective structure,
- one central controller-like locus of integration.

The problem

Embodied and biological systems often look more like coordination across partially independent processes than command from one central unit (Shaw 2026a).

Polyphonic active inference: the intuition

Polyphonic active inference keeps multiple inferential “voices” active at once.

- A voice may encode goal pursuit, safety, epistemic sampling, stability, memory or social/contextual constraint.
- Voices can disagree without immediately collapsing into a single winner.
- Behaviour emerges from constrained coordination among voices.

Polyphonic does not mean chaotic

It means structured pluralism: multiple partially compatible pressures remain visible and interpretable - and contribute to policy decisions (Shaw 2026b).

Polyphonic active inference: multiple voices

$$F_i[q_i] = \mathbb{E}_{q_i(s)} [\ln q_i(s) - \ln p_i(o, s)]$$

i	One inferential voice or subsystem.
$q_i(s)$	That voice's current belief about hidden states.
$p_i(o, s)$	That voice's generative model of observations and causes.
$F_i[q_i]$	The free energy minimised by that voice.

Each voice explains the world from its own perspective.

Polyphonic action: integration rather than command

$$G_i(\pi) = \text{risk}_i + \text{ambiguity}_i - \text{epistemic value}_i$$

risk_i

How far the imagined future is from what voice i prefers.

ambiguity_i

How unclear or unreliable the future observations are expected to be.

epistemic value_i

How much the policy is expected to teach that voice.

$$u = \sum_i w_i u_i$$

Each voice proposes an action tendency u_i . Behaviour is the weighted integration of these tendencies, not a winner-takes-all command.

Polyphonic influence weights

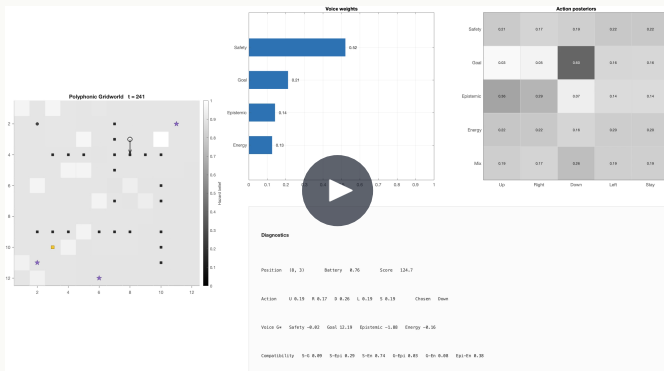
$$w_i = \sigma \left(-\gamma_i G_i + \sum_{j \neq i} K_{ij} \phi_{ij} \right)$$

i	Inferential voice or subsystem.
u_i	Action tendency proposed by voice i .
w_i	Current influence of voice i .
K_{ij}	Compatibility or coupling between voices i and j .
ϕ_{ij}	Evidence that voices support, conflict or constrain each other.

Polyphonic gridworld: seeing the voices

What the video adds

- The agent is not driven by one scalar objective.
- Safety, goal, epistemic and energy voices retain separable influence.
- The mixed action posterior makes coordination inspectable.



Click thumbnail to open the included video.

Coordination without command

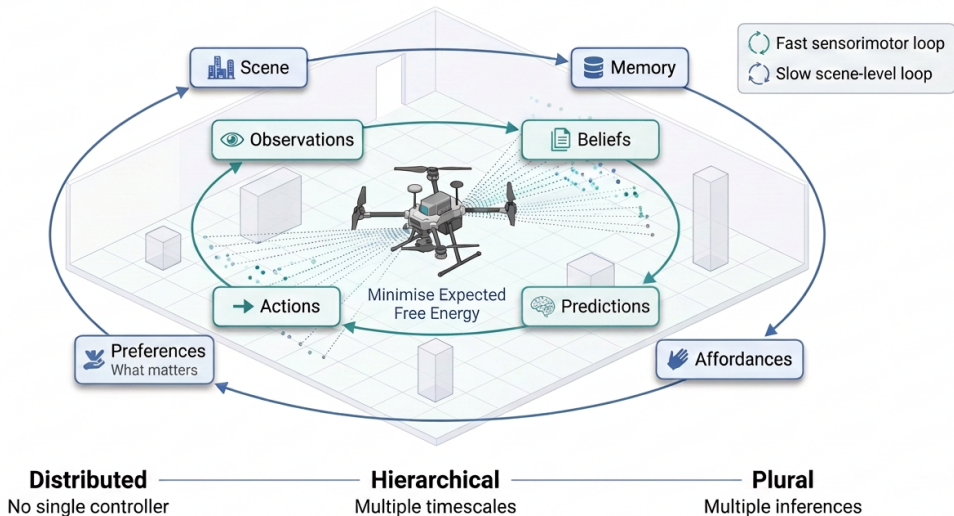
- In embodied agents, the body is part of the inference problem, not just an output device.
- Local sensorimotor loops may handle fast contingencies while slower contextual processes shape what matters.
- The key design question shifts from *who is in control?* to *how are multiple inferential processes coordinated?*

Preprint argument

Robust embodied behaviour may depend less on exhaustive central resolution than on coordination across timescales, interfaces and inferential demands (Shaw 2026a).

Coordination Without Command

Active inference and the route to embodied intelligence



Drone example: active inference under partial observability

- The drone is not given simulator truth; it receives noisy self cues, egocentric target cues and ray-based obstacle sensing.
- Fast beliefs track self state, target state and local geometry.
- Slower scene beliefs track visibility, progress, affordance, memory and context.
- Candidate policies are evaluated for target progress, safety, stability, epistemic value and future scene improvement.

Why it matters

The controller can remember a target under occlusion, distinguish blockage from poor vantage and choose temporarily indirect actions to improve future evidence.

Drone demo: what we built

Physical world

- A PyBullet 3D environment with gravity, collisions and continuous movement.
- A drone body that must move through space rather than jump between abstract states.
- Observations are partial and noisy, so the controller does not receive perfect simulator truth.

Belief system

- The drone maintains beliefs about where it is and what is happening around it.
- It predicts what its actions should make it observe next.
- When observations differ from predictions, beliefs and actions are updated.

sense → infer → predict → act → update

Drone demo: active inference in PyBullet

What to point out

- The drone acts from its own inferred state, not from privileged access to the full simulation.
- Control emerges from repeatedly reducing the mismatch between predicted and observed movement.
- Behaviour looks like correction and recovery, not a pre-scripted animation.



World models: semantic priors

- The agent is not searching randomly.
- It is equipped with a semantic prior: cups are more likely to be in some places than others.
- Before seeing anything, the kitchen is therefore a sensible first hypothesis.
- Active inference lets the agent combine this prior knowledge with new sensory evidence.

semantic prior → search policy → new evidence → updated belief

The world model contains structured expectations about objects, places and actions, not just coordinates.

World model demo: looking for the cup

What the video shows

- The agent starts with a prior belief that the cup is probably in the kitchen.
- It searches the kitchen first because that is semantically plausible.
- If the cup is not found, this negative evidence reduces belief in that location.
- The agent then redirects search to the next plausible place.



Take-home points

1. Active inference starts with a generative model: hidden states cause observations.
2. Perception minimises variational free energy by updating beliefs.
3. Action minimises expected free energy by selecting informative and preferred futures.
4. Precision determines which errors, preferences and voices dominate behaviour.
5. Polyphonic active inference keeps multiple inferential pressures visible rather than collapsing everything into one command centre.

Thank you

Questions?

a.d.shaw@exeter.ac.uk | cpnslab.com

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